Sanat Kumar Agrawal

Final Year Dual Degree Student at IIT Bombay

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Portfolio
LinkedIn

Education

- Indian Institute of Technology, Bombay (Oct '22-Present)
  - Dual Degree (B.Tech.+M.Tech.), **Electrical Engg.**, Minor in **Machine Intelligence & Data Science** **CPI: 9.54/10**
- École Polytechnique Fédérale de Lausanne (EPFL), Switzerland (Sept '25-Jan '26)
  - Semester Exchange in **Electrical Engineering**, funded by the Heyning-Roelli Scholarship **Grade: 5.61/6**

Publications

- Karl El Hajal[†], **Sanat Kumar Agrawal**[†], Enno Hermann, Mathew Magimai-Doss. "*Concatenative Neural Speech Synthesis.*" Accepted at **IEEE SLT 2026** (poster). [†]Equal contribution. [Demo] *(Sept '26)*
- **Sanat Kumar Agrawal**, Srikanth Raj Chetupalli. "*Indic-CLAP: Text-Audio Understanding in Indic Languages via Cross-Lingual Distillation.*" Under review at **ICASSP 2027**. [arXiv] [Demo] *(Sept '26)*
- **Sanat Kumar Agrawal**, Srikanth Raj Chetupalli. "*Timestamped Hindi Speech Transcription Using Whisper.*" Under review at **ICASSP 2027**. [arXiv] [Demo, Model] *(Sept '26)*

Research Experience

Concatenative Neural Speech Synthesis | Research Internship (Oct '25 - Feb '26)

Guide: **Dr. Mathew Magimai-Doss, Idiap Research Institute, Switzerland**

- Proposed a **training-free** TTS method for new domains that concatenates speech units in **WavLM** feature space.
- Replaced the text-to-SSL model of **kNN-TTS** with unit concatenation, removing all supervised text-audio training.
- Showed that feature-space concatenation improves **naturalness** over Festival TTS, at some cost in intelligibility.
- Ran a six-configuration **ablation** isolating the effects of SSL-space concatenation and kNN-VC retrieval on quality.
- Remained competitive with **XTTS** on zero-shot speaker similarity with only **0.9 hours** of speech versus its 27,282.

Indic-CLAP: Cross-Lingual Text-Audio Retrieval | Undergraduate Research (Mar '26 - Present)

Guide: **Prof. Srikanth Raj Chetupalli, Digital Audio Processing Lab, IIT Bombay**

- Built **Indic-CLAP**, extending CLAP to 9 Indic languages by distilling a multilingual text encoder from frozen CLAP.
- Proposed a **hybrid distillation** loss reaching **38.8% A→T R@1**, beating the English CLAP teacher on text retrieval.
- Compared 4 distillation objectives, finding text targets aid zero-shot classification and audio targets aid retrieval.
- Explained the gains with a **modality gap** analysis, and showed that the recipe transfers to frozen GLAP encoders.
- Traced the gap between Indo-Aryan and Dravidian languages to translation quality, corpus size and token length.

Timestamped Hindi Speech Transcription | Undergraduate Research (Mar '26 - Present)

Guide: **Prof. Srikanth Raj Chetupalli, Digital Audio Processing Lab, IIT Bombay**

- Trained a character-level **CTC head** on the Whisper encoder for Hindi word timestamps, comparable to WhisperX.
- Kept the Whisper model frozen and trained only a lightweight head, avoiding a second alignment model entirely.
- Used the CTC output to curb **hallucinations**, reducing WER from 39.1% to 32.6% on conversational DISPLACE-M.
- Released the model on **Hugging Face** along with an interactive demo showing predicted Hindi word boundaries.

Parametric Uncertainty Estimation in DAB Converters | SURI Internship (May '24 - Jul '24)

Guide: **Prof. Ayan Mallik, PEACE Lab, Arizona State University, USA**

- Built an all-loss-inclusive **MATLAB** model of a dual active bridge (DAB) converter to estimate aging-induced drift.
- Moved simulation to **PLECS** with parallel runs, scaling data generation by over **30**×, from 230 to 6,000+ samples.
- Designed physics-based features and a **PINN** with **0.2%** inductance error, matching a plain NN with half the data.
- Calibrated a converter model on 12 hardware measurements to generate 720K samples for multi-output training.

Industry Experience

Founding Engineer | **Novare Talent** (Aug '25 - Present)
AI-native hiring platform, incubated at SINE, IIT Bombay | Backed by NVIDIA Inception
- Shipped 3 production ML systems on **FastAPI**, Next.js and Supabase, serving **2,300+ users** from 14 IIT campuses.
- Engineered **pgvector** semantic search with **LLM reranking**, plus a WhatsApp agent that returns ranked shortlists.
- Built **Arena X**, a real-time 1v1 coding arena with ELO matchmaking over sandboxed test-case judging for hiring.

AI & Systems Intern | **SalesAgent.ai** (Jun '25 - Jul '25)
Speech pipeline for multilingual conversational AI
- Built a streaming **faster-whisper** ASR server for 8 kHz Hindi-English telephony with runtime language switching.
- Added confidence-based **hallucination gating** and silence-based endpointing for natural turn-taking on live calls.
- Profiled the pipeline end to end and found **99.9%** of latency was model inference, which redirected optimisation.

Key Projects

Autonomous Racing Agent via Reinforcement Learning [Code] (Apr '25 - May '25)
Guide: **Prof. Shivaram Kalyanakrishnan** | CS-747: Foundations of Intelligent and Learning Agents
- Engineered 3 core features from a 13×13 sensor grid, cutting input dimensions by **over 95%** for real-time control.
- Trained a hybrid policy using **CMA-ES** over 100 generations, achieving **3.2**$\times$ the distance with 80% fewer crashes.
- Evolved a 30-agent population per generation, reaching consistent driving across dynamic race tracks and turns.

LLM Decoding and Inference Acceleration with Medusa (Apr '25 - May '25)
Guide: **Prof. Sunita Sarawagi** | CS-726: Advanced Machine Learning
- Implemented greedy, top-k, nucleus and temperature decoding on **Llama-2** for translating Hindi to English text.
- Built trie-based word-constrained and **Medusa-style speculative decoding**, achieving up to **3**$\times$ faster inference.
- Benchmarked all strategies on the **IN22-Gen** set with BLEU and ROUGE to quantify their speed-quality trade-off.

Reverse-Mode Automatic Differentiation Engine [Code] (Sept '25 - Dec '25)
CS-328: Numerical Methods for Visual Computing & ML | EPFL, Switzerland
- Built a reverse-mode **autodiff** engine from scratch and used it to train neural networks without an ML framework.
- Implemented SVD and PCA face recognition, and **QR**-based least-squares solvers for image deblurring problems.

Capacitive Coupling in 2D Field-Effect Transistors [Code] (Sept '25 - Dec '25)
Physical Models for Micro and Nanosystems | EPFL, Switzerland
- Modelled a nanoscale **2D-material FET** in COMSOL, solving Poisson's equation for the electrostatic potential field.
- Extracted gate-channel capacitances by energy methods, studying how gate geometry scales inter-gate coupling.

Cough-Based Disease Detection using ML [Code] (Jan '25 - May '25)
Guide: **Prof. Nirmal Punjabi** | DH-307: R&D Project
- Built a dual-input **BiLSTM-CNN** reaching **96.33%** tuberculosis accuracy and 94% for COVID-19 from cough audio.
- Raised F1 by **14%** on imbalanced Coughvid and Coswara data with SMOTE, and cut features by 89% using MRMR.

16-bit Pipelined RISC Processor [Code] (Apr '24 - May '24)
Guide: **Prof. Virendra Singh** | Microprocessors
- Designed a 16-bit **pipelined RISC** CPU in VHDL with a 6-stage datapath and five pipeline registers between stages.
- Added data forwarding for hazards, predicated execution and load/store-multiple, tested via a custom testbench.

Smart Walker for Clinical Rehabilitation [Code] (Jan '25 - Apr '25)
Electronic Design Lab | IIT Bombay
- Prototyped a **Raspberry Pi** walker with load cells, a custom PCB and CAD enclosures for gait and weight analysis.
- Built **ArUco**-marker foot tracking that avoids the occlusion failures of YOLO and OpenPose, with haptic feedback.
- Resolved an **SPI** pin conflict between the load cells and LED feedback on the Raspberry Pi using PIO reallocation.

AlphaSense AI: Live Equity Trading System | **Self Project** [Code] (Apr '26 - Aug '26)
Selected for the IITB-Groww INV.ENT Program
- Fused LLM news sentiment, earnings-call audio analysis and XBRL fundamentals into 7 strategies for NSE stocks.
- Validated strategies via **walk-forward** backtests, dropped 2 that failed out-of-sample, and ran them live on AWS.
- Scored earnings calls for **management confidence** using hedging, repetition and disfluency cues from each call.

FitSurv: Real-Time Form Correction | **Self Project** [Code] (2025)
- Built a real-time posture and rep-counting coach with **MediaPipe** pose tracking and joint-angle analysis for form.
- Ran under **0.1 s** latency at over 90% accuracy; the venture won a **Rs 2.5 lakh** grant under IDEAS L1 at IIT Bombay.

Scholastic Achievements

- Scored a perfect **10.0/10 Semester Performance Index** in the Spring 2025 semester at IIT Bombay (2025)
- Awarded the **Heyning-Roelli Scholarship** to pursue an academic exchange semester at EPFL, Switzerland (2025)
- Awarded a **Rs 2.5 lakh** grant under **IDEAS L1**, IIT Bombay's flagship pre-incubation program run by DSSE (2025)
- Selected globally for the fully funded **Summer Research Initiative (SURI)** at Arizona State University (2024)
- Secured an All India Rank of **826** in IIT-JEE Advanced examination among over **0.15 million** candidates (2022)
- Secured an All India Rank of **527** in JEE Main examination among over **0.9 million** candidates (2022)

Technical Skills

Languages & Tools: Python, C++, Java, SQL, VHDL, Bash, MATLAB, Simulink, PLECS, COMSOL, LTSpice, KiCad, LaTeX

ML & Speech: PyTorch, HuggingFace, Scikit-learn, TensorFlow, Keras, faster-whisper, WavLM, Festival, CUDA, Slurm

Systems: FastAPI, Next.js, React, Node.js, Supabase, PostgreSQL, pgvector, AWS EC2, Docker, Git, Linux, Streamlit

Teaching Experience

Teaching Assistant | **EE-103** | **ENT-609**
- EE-103: **Introduction to Electrical Engineering** | Tutorials and grading for **200+** students (Aug '26 - Present)
- ENT-609: **Marketing and Finance for Entrepreneurs** | Lectures and grading for **40+** students (Aug '24 - Nov '24)

Key Courses

ML & Speech: Advanced Machine Learning, Intelligent & Learning Agents, Introduction to ML, Data Science and ML
Signals & Systems: Digital Signal Processing, Error Correcting Codes, Communication Networks, Microprocessors
Mathematics: Linear Algebra, Probability & Random Processes, Markov Chains, Online Learning and Optimisation
EPFL: Speech Processing, Numerical Methods for Visual Computing & ML, Intro to Quantum Information Processing

Positions of Responsibility

Web Coordinator | **Student Alumni Relations Cell, IIT Bombay** (Jun '23 - Apr '24)
- Worked in a core team of **80+** members, organizing **40+** events that connect **60k+** alumni with current students.
- Built the portal for mentors and mentees of the Alumni-Student Mentorship Programme in Django and Bootstrap.

Extracurricular Activities

- Conducted an **8-session AI literacy workshop**, teaching core AI concepts and tools to 15+ school students online.
- Backpacked solo across **15+ European countries** during the exchange semester, planning every trip on my own.
- Completed a yearlong intensive **chess** training programme under the National Sports Organisation, IIT Bombay.
- Conducted **doubt-solving sessions** for underprivileged school students in collaboration with IIT Kanpur in 2023.

References

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Prof. Ayan Mallik
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